

Know When the Epidemic Comes: Mathematics Before Data in Dengue Outbreak Forecasting

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ABSTRACT

Early warning asks *when* an epidemic arrives, not only how many cases follow. We show these are different problems on district-level dengue, and that mechanistic structure answers the first while data answers the second. Spatio-temporal graph neural networks are widely applied here, yet their advantage over naive baselines is rarely verified under a leak-free protocol. We reproduce five published ST-GNNs on Sri Lanka’s dengue record (25 districts, 459 weeks) to within 7.3%, and find the reported gains are measured with training data included: naive persistence beats every model in the column the source paper reports. We also identify an administrative reporting backlog occupying one fold’s test split and carrying $\sim 90\%$ of its squared error; scoring with and without those windows moves the persistence floor from RMSE 44.80 to 29.52 and erases the best architecture’s 3.8-point lead. Against that floor, no architecture and none of four evidence-led increments separate from their own baseline at $p < 0.05$. Testing the mechanistic prior the task invites—a renewal equation whose generation interval comes from a validated SEIR–SEI model—it reconstructs the record given oracle R_t (RMSE 13.44) and fails as a forecaster, because R_t is only 26% predictable, so routing a forecast through it re-injects a $1.90\times$ error per step where persistence carries none. The ceiling is informational, not architectural. The mechanism is nonetheless useful in the two roles that do not ask it to predict: as an inequality constraint it reliably improves a model (-0.044 RMSE on 57/60 paired runs, $p < 0.0001$), and it is the largest single contributor to ranking which districts will breach an alert threshold (AUC $0.807 \rightarrow 0.826$). Mechanistic knowledge pays here when it restricts the hypothesis space, not when it supplies a forecast.

KEYWORDS

dengue forecasting, graph neural networks, reproducibility, evaluation protocols, physics-informed learning, epidemic early warning, negative results

1 INTRODUCTION

Dengue is the fastest-spreading mosquito-borne viral disease globally, and Sri Lanka faces recurring severe epidemics: the 2017 outbreak alone recorded over 186,000 cases and 440 deaths [5]. Roughly

three weeks separate infection from clinical diagnosis, so district-level forecasts issued $H = 3$ weeks ahead carry actionable lead time for pre-positioning clinical capacity and intensifying vector control.

Three properties shape the task. Transmission is spatially coupled, which motivates a graph; the panel is small (25 districts, 459 weeks) so parameters are cheap to overfit; and the target is severely heavy-tailed (skewness 8.6, the top 1% of district-weeks holding 18.3% of cases), which makes squared error a measurement of a few dozen windows.

A fourth property is easy to miss and dominates the rest: weekly incidence is strongly autocorrelated. Cases at $t - 1$ explain $r^2 = 0.85$ of cases at t . Any model is therefore competing against a naive persistence forecast $\hat{y}_{t+h} = y_t$ that is already close to optimal, and a benchmark that omits that baseline cannot tell whether its architecture contributed anything. Weng et al. [6], the reference benchmark for this dataset, omit it.

We proposed three stackable contributions to an ST-GNN baseline: (a) a physics-informed loss from an SEIR–SEI compartmental model, as $\mathcal{L}_{\text{data}} + \lambda_p \mathcal{L}_{\text{phys}} + \lambda_c \mathcal{L}_{\text{cons}} + \lambda_s \mathcal{L}_{\text{smooth}}$; (b) GAN augmentation; and (c) a learned adaptive graph. This paper reports (a) and (c); (b) is left to the full study. Establishing what the baseline actually *was* turned out to be the harder problem, and it reframes the target. Cases at $t - 1$ explain $r^2 = 0.85$ of cases at t , so *how many* is close to answered before a model starts; *when*—which district crosses an alert threshold, and roughly *when*—is not. That is the question early warning actually asks, and it is where we find the mechanism earns its place.

Contributions.

- **Exact reproduction.** Five published ST-GNNs to within 7.3% across twenty comparisons; the reported column is training-inclusive and per-window averaged, and naive persistence beats every model in it (§4).
- **An artifact-aware protocol.** A $19\times$ reporting backlog at week 395 occupies one fold’s test split, carries $\sim 90\%$ of its squared error and inverts fold difficulty; we score every arm and the baseline with and without it (§3).
- **A null result, stated as one.** Over 3 origins $\times$ 3 seeds, no architecture and no increment differs from its own baseline at $p < 0.05$ against an artifact-free floor of 29.52 (§5).

- **Why the mechanistic prior cannot help, and where it can.** R_t is 26% predictable, implying a $1.90\times$ error per step (§6); the same quantity is the largest single contributor to ranking outbreaks three weeks ahead (§6.2).

2 RELATED WORK

ST-GNNs for epidemic forecasting. Spatio-temporal graph networks couple a graph convolution over regions with a temporal operator over history. The architectures we evaluate—STGAT, A3TGCN, ASTGCN, DCRNN and AAGCN—originate in traffic forecasting and were transferred to dengue by Weng et al. [6], who report them on the dataset used here. GulMohamed et al. [1] propose per-horizon output layers, additive temporal attention and a Gaussian predictive head; we adopt all three as increments and test them (§5).

Mechanistic models and hybrids. Compartmental models remain the standard description of dengue transmission; we use the SEIR–SEI formulation and sensitivity analysis of Phaijoo and Gurung [3], whose nine published sensitivity indices we reproduce. Physics-informed neural networks embed such dynamics as a loss term [4], and recent work applies them to epidemic curves, typically in a regime where data are scarce relative to the dynamics—MP-PINN [2] trains on 35 days and forecasts 97. That is the regime where a mechanistic prior pays. Our setting is the opposite: 459 weeks across 25 districts forecasting three weeks, where a naive baseline is already near-optimal. We show that the difference is not incidental but decides the outcome, and we quantify the crossing point in terms of the predictability of the reproduction number.

3 DATA AND EVALUATION PROTOCOL

3.1 Data

We use the processed national surveillance array of Weng et al. [6]: 459 weekly observations for Sri Lanka’s 25 districts (2013–2022), weekly reported cases as the target, ten lag-shifted climate covariates, and a district graph of 141 directed edges with self-loops. The target is heavy-tailed (median 13, maximum 2631, 9.7% zeros, skewness 8.6), so we model $\log(1 + y)$. The dominant statistical fact is temporal, not spatial: cases at $t - 1$ explain $r^2 = 0.85$ of cases at t , while the best single covariate explains 0.02. Every number below should be read against that.

3.2 Protocol

We use rolling-origin (expanding-window) cross-validation with three chronological origins (0.55, 0.70, 0.85 of the series) and three seeds, a $W = 3$ week input window and an $H = 3$ week horizon. Normalisation statistics are computed from training weeks only and recomputed per fold; no shuffling occurs at any point. Model selection uses a genuine validation split of the final 30 training weeks, and we report *pooled* RMSE over held-out windows.

Three of those choices are corrections to the reference protocol rather than preferences, and §4 quantifies each.

3.3 An evaluation artifact, and dual scoring

Exploratory analysis identified week 395 as a $19\times$ spike in reported cases occurring simultaneously in 18 of 25 districts. A simultaneous

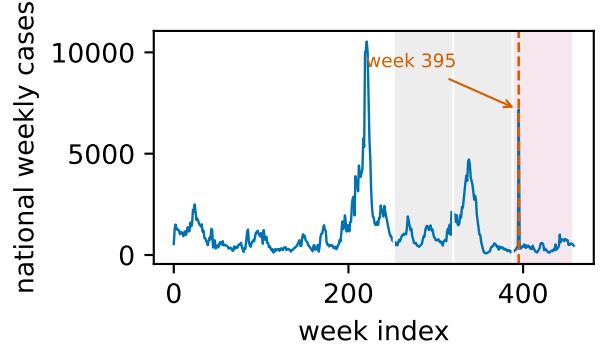


Figure 1: The national series, with the three test splits shaded. The week-395 backlog lands in the last one. On that fold persistence scores RMSE 68.62 over all 68 windows, 219.07 over the six touching week 395, and 22.80 over the remaining 62.

Table 1: The mechanistic route, measured. Every anchor available at forecast time is worse than persistence, and every candidate predictor of transmission intensity explains almost nothing out of sample.

Renewal anchors — artifact-free RMSE	
persistence	29.52
$R=1$ (force only)	35.45
ratio form, damped	33.05
$\hat{R}$ -force	50.01
own past $\rightarrow \log R_t$ (r^2 , out of sample)	0.263
climate $\rightarrow \log R_t$	-0.025

multi-district jump of that magnitude is an administrative reporting backlog, not an epidemiological event.

Under our protocol it falls in the *test* split of the origin-0.85 fold and in no training split at all. Six of that fold’s 68 windows touch it, and they carry approximately 90% of its squared error (Fig. 1). Persistence scores RMSE 68.62 over all 68 windows, 219.07 over the six, and 22.80 over the remaining 62.

This has two consequences that any benchmark on this dataset inherits. First, the fold ordering inverts: measured over all windows, origin 0.85 is by far the hardest fold (68.62 against 26.88 and 38.89); measured artifact-free it is the *easiest* (22.80). A claim about “the hard fold” is therefore meaningless without stating which scoring it came from. Second, because that fold dominates the pooled average, the persistence floor itself moves—from RMSE 44.80 to 29.52—and so does every model’s apparent margin against it.

We therefore score every arm twice: over all test windows, and over the artifact-free subset. Crucially the *baseline* is *split the same way*. Scoring a model artifact-free against a floor computed over all windows compares two different test sets. We do not drop the windows silently, because doing so flatters every arm equally and conceals that a fold’s difficulty was one week of bad data.

4 REPRODUCING THE BENCHMARK

We ran the reference implementation [6] unmodified—their code, their data, their protocol—on a pinned software stack, and recovered their published numbers to within 7.3% across twenty comparisons, with AAGCN and DCRNN matching to four decimal places. Three deviations from their environment were forced and are documented with the exact error that forces each; none changes a reported figure. The published results are real.

What they measure is a different question, and three properties of the reference evaluation account for the entire apparent advantage.

The reported column includes training data. The column labelled “cross validated” is computed on a loader spanning the full series rather than a held-out split; it is a fit statistic, and the column names are effectively inverted relative to their meanings. **RMSE is averaged per window, not pooled.** By Jensen’s inequality that is systematically lower, and on this target the gap is ~39%. **The naive baseline is absent.** Adding persistence to their own reported column reverses the result: it beats *every* model there, on both MAE and RMSE. An independent re-implementation sharing no code with either codebase reproduces this, so it is not an artefact of one.

Published advantages here therefore do not survive contact with a naive baseline on held-out data. We take only the *architectures* forward, under §3; numbers below are comparable to each other, not to the source table.

5 ARCHITECTURES AND INCREMENTS

We evaluate the five reproduced architectures under the protocol of §3, each with four increments justified by prior work or by our own exploratory analysis: per-horizon output heads, additive temporal attention and a Gaussian predictive head [1], and a Huber objective motivated by the target’s skewness. All five architectures accept all four increments through a shared output seam, so no cell of the table covers an unstated subset.

Figure 2 gives the result. Measured over all windows, AAGCN (40.99) appears to beat the 44.80 floor by 3.8 points and ASTGCN by 1.4. Measured artifact-free, AAGCN sits at 29.84 against a floor of 29.52—*above* it—and the ordering of architectures changes. The apparent winner was the architecture that best absorbed six windows of backlogged reporting.

Artifact-free, only STGAT (29.35) and A3TGCN (29.05) are below the floor at all, by 0.6% and 1.6%. Under paired *t*-tests on matched (architecture, origin, seed) runs, **no increment differs from its own baseline at $p < 0.05$:** per-horizon heads +0.526 ($p = 0.078$), temporal attention +0.245 ($p = 0.077$), Huber +0.128 ($p = 0.462$), Gaussian head +0.280 ($p = 0.304$). The two smallest *p*-values belong to increments that are *worse* than baseline. We report this as the null it is; selecting the minimum of a table this noisy would manufacture a result.

One increment does earn its place, on a different axis. The Gaussian head yields 95% prediction intervals with empirical coverage between 0.947 and 0.962 across all five architectures—calibrated intervals that a point forecast cannot supply, and the operationally useful output of the whole study.

6 THE MECHANISTIC PRIOR

The project’s intended contribution was a physics-informed loss from the SEIR–SEI model. A hard compartmental residual is not identifiable here—seven compartments with exactly one observed, and that one under-ascertained—so we use the mechanistic identity that acts on the observable, the renewal equation

$$y_t = R_t \sum_{s \geq 1} w_s y_{t-s}, \quad (1)$$

where the generation-interval distribution w follows from the SEIR–SEI stage durations of Phaijoo and Gurung [3], whose nine sensitivity indices we reproduce. The implied mean generation interval is 3.30 weeks. Only the stage durations are used; their R_0 is not, since generation interval is disease biology and transmission intensity is local.

Equation (1) describes the record well. Given oracle R_t , a three-week replay scores RMSE 13.44 where persistence scores 55.03 on the same windows. We tested it both as a soft penalty and as a decoder in which the network emits $\log R_t$ and cases are reconstructed through the mechanism. The decoder does exactly what it was built to do—maximum predicted weekly log-growth rises from 0.07 to 4.14, so the flatness is gone—and is 6 RMSE worse anyway. The penalty degrades RMSE monotonically in its weight.

Table 1 localises the failure. Every causally available anchor loses to persistence: force alone ($R=1$) scores 35.45, a damped ratio form 33.05, and $\hat{R} \cdot$ force 50.01, against 29.52. The predictive content of Eq. (1) is entirely in R_t —and R_t is only 26% predictable out of sample, from its own past and from nothing else. With $\text{sd}(\log R) = 0.749$, that leaves a residual multiplicative error of $\exp(0.64) = 1.90\times$ per step, compounding over three weeks. Persistence carries no such factor. *Reconstructing a forecast through R re-injects a factor-of-two error where the naive baseline had none.*

Neither mechanism that could supply the missing predictability does. **Susceptible depletion** adds nothing beyond R ’s own past (r^2 0.263 $\rightarrow$ 0.263): even at 1-in-20 under-ascertainment, cumulative infection over 8.8 years moves $\log R$ by 0.618, about 0.0013 per week against a weekly sd of 0.749. Sri Lankan dengue is endemic in 21.9M people who do not run out of susceptibles on a three-week horizon. **Climate driving R_0** fails too: the sensitivity indices ($b + 1.000$, $\mu_v - 0.818$, $m + 0.500$, $v_v + 0.318$) imply a hump-shaped thermal response, and allowing that curvature moves out-of-sample r^2 from -0.025 to -0.028 . We did not fit entomological curves to our own data: a constant estimated from the series it then constrains is not a prior.

Table 1 also gives the sign error directly: $\log \hat{R}_t$ correlates +0.798 with the *past* three weeks of growth and -0.217 with the next three. It is a backward-looking state estimate, so a penalty pulling a forecast toward it pulls the wrong way—which is the monotonic damage we measured. A physics-informed *variance* term fails for the same reason and with the same sign: $\log \hat{R}_t$ correlates -0.106 with the magnitude of future growth, so elevated $\hat{R}$ does not even signal elevated uncertainty.

6.1 The ceiling is informational

Regressing each model’s predicted log-growth on realised log-growth gives a slope of **0.002**. The models’ growth predictions

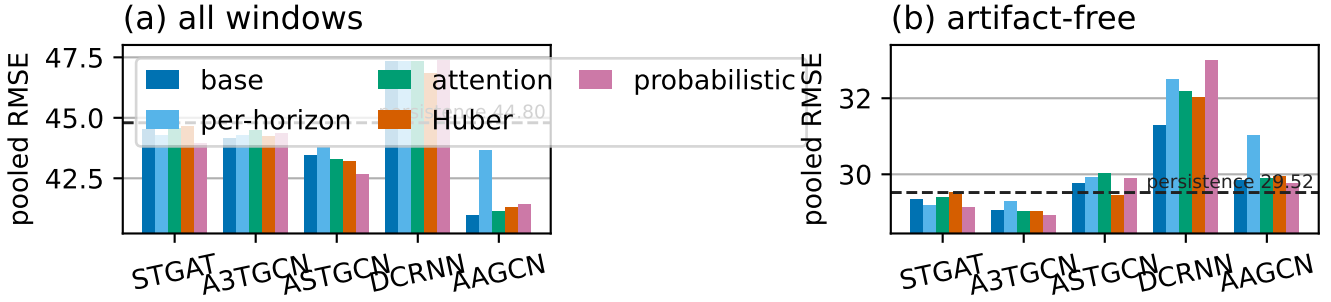


Figure 2: Every arm against the persistence floor, scored both ways. AAGCN’s 3.8-point lead in (a) does not survive in (b): it was a lead at predicting a reporting backlog. Bars are means over 3 origins $\times$ 3 seeds.

carry no information about growth; they are persistence with noise. Under squared error this is *correct*: the target is a residual over a baseline that already explains $r^2 = 0.85$, and the conditional mean of what remains is approximately zero. Flatness is what the objective asks for. This is why an asymmetric loss, the renewal penalty and the renewal decoder each bought responsiveness and paid for it in RMSE, in the same direction: weighting $3\times$ against under-prediction removes the outbreak bias entirely ($-10.58 \rightarrow +1.21$ cases) and costs 2.8 RMSE, because the model cannot tell which windows are outbreaks.

6.2 What the mechanism can still do: constrain, not reconstruct

Every failure above shares a form. The decoder routes the forecast *through* R ; the penalty pulls it *toward* a target derived from $\hat{R}$. Both make the mechanism carry predictive weight, which $\hat{R}$ cannot. An *inequality* constraint does not: it is zero inside the admissible region and cannot inject a target. We test two, added to the reproduced architectures’ objective and scored against each architecture’s own base on matched (origin, seed) runs at 20–25 seeds. A **biological envelope** bounds weekly log-growth above by $r_{\max} = 2.3884$ and below by the host clearance rate. A **spatial term** penalises disagreement in *relative* incidence $\hat{y}_i/s_i$ between adjacent districts—the scale-free form of a graph-Laplacian penalty, which on raw counts merely taxes population.

The two separate cleanly, and only at adequate n . The **spatial term holds**: on STGAT it improves artifact-free RMSE by -0.044 on 57/60 matched runs ($p < 0.0001$), and independently on each origin ($p = 0.040$, < 0.0001 , 0.025). The **envelope does not**: -0.186 with 5/9 better at $n = 9$ becomes -0.042 with 38/75 at $n = 75$ ($p = 0.69$), a coin flip. At $n = 9$ the two were indistinguishable. The envelope is in any case *provably inert* wherever a model under-reacts—STGAT and A3TGCN peak at 0.31 against a ceiling of 2.3884 and their envelope arms are bit-identical to base on every run—which restates EXP-014 as a design rule: a bound helps exactly the models that violate it, and here most do not.

The mechanism’s second use is a different target. Ranking—which districts will breach an alert threshold, rather than by how much—is the operationally relevant question and a tractable one.

Defining an outbreak as a district-week above that district’s training-set 90th percentile within the horizon (base rate 14.4%), current level alone reaches AUC **0.807**; adding $\hat{R}$ raises it to **0.826**, a larger gain than recent growth ($+0.004$) or neighbouring districts ($+0.006$), while climate again subtracts. Before onset, mean $\hat{R}$ runs 1.13–1.24 for six weeks against a global median of 0.911—sustained $R > 1$, visible before the outbreak is.

The forecasting gain is real and small: 0.15% of a 29.52 floor. It is a reliable improvement *of the model*, not a move past the baseline—STGAT goes from 29.550 to 29.506 against a floor of 29.52, a margin that remains inside noise ($p = 0.70$). So §6 stands, with one qualification we think is the useful part. On a target a naive baseline already explains to $r^2 = 0.85$, mechanistic knowledge pays when it *restricts the hypothesis space* and not when it supplies a prediction: a constraint, not a forecast. Only the first survives a poorly-determined R_t .

7 DISCUSSION AND CONCLUSION

When does a mechanistic prior pay? PINNs succeed where data are scarce relative to the dynamics—MP-PINN [2] trains on 35 days and forecasts 97. Ours is the opposite regime, and the decisive quantity is not sample size but how much a naive baseline already explains: here $r^2 = 0.85$, leaving an R_t that is 26% predictable. That ratio predicts *which* use survives—a prior supplying a forecast inherits R_t ’s error, one restricting the hypothesis space does not.

Reporting. A caution that applies to us as much as to the work we reproduce: at $n = 9$ the spatial constraint ($p = 0.057$) and the growth envelope ($p = 0.238$) were indistinguishable; at $n \geq 60$ one holds and the other is a coin flip. Small effects on folds this noisy need the seeds to separate them.

Limitations. One country, one disease, one processed array. We do not evaluate GAN augmentation—the remaining proposed contribution—and the detection result uses a single-week threshold where sustained exceedance would be more operationally meaningful.

Conclusion. Under a leak-free, artifact-aware protocol no reproduced architecture and no increment separates from a persistence floor of 29.52, and reconstructing a forecast through the mechanism cannot close that gap. But the mechanism holds in the two roles that do not ask it to predict—as an inequality constraint and as

a state estimate for ranking outbreaks. Knowing *when* is where the mathematics comes first. Code and a reproduction notebook accompany this paper.

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